Rust Tutorial: Ownership and Borrowing

Preparation document

Florian Walter and Anto Grgic · Programming Languages and Paradigms, UZH · 11.11.2026

1 Install Rust

Rust is usually installed with `rustup`. It installs the Rust compiler, called `rustc`, and the build tool `cargo`. It also keeps Rust up to date.

You can find the official instructions at https://www.rust-lang.org/tools/install. Other installation options are listed at https://forge.rust-lang.org/infra/other-installation-methods.html.

Windows

Download and run `rustup-init.exe` from https://win.rustup.rs/. The standard Windows version of Rust also needs the Microsoft C++ Build Tools or Visual Studio. The installer can help you set these up. More details are available at https://rust-lang.github.io/rustup/installation/windows-msvc.html.

You can also install WSL and then follow the Linux instructions below. The WSL setup guide is at https://learn.microsoft.com/en-us/windows/wsl/install.

macOS

First install the Xcode Command Line Tools. Rust needs them for the linker. Then install Rust with `rustup`.

```
xcode-select --install
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
```

You can also use Homebrew. Run `brew install rustup`, followed by `rustup default stable`. You can get Homebrew from https://brew.sh/.

Linux

You need a C compiler and linker before installing Rust. On Debian or Ubuntu, run `sudo apt install build-essential`. On other distributions, install a package such as `gcc`. Then run:

```
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
```

We recommend `rustup` instead of your distribution's Rust package, which may be out of date. Restart your terminal after the installation. You can also run `source $HOME/.cargo/env` to make `cargo` available right away.

2 Choose an editor

Use any editor you are comfortable with. We recommend one that supports **rust-analyzer**, since it shows Rust errors while you type. See https://rust-analyzer.github.io/ for more information.

Some good options are:

- VS Code with the rust-analyzer extension. This is probably the simplest setup. Download VS Code at https://code.visualstudio.com/ and the extension at https://marketplace.visualstudio.com/items?itemName=rust-lang.rust-analyzer. Do not install the old extension named "Rust".
- Neovim or Helix with rust-analyzer set up as the language server.
- RustRover from https://www.jetbrains.com/rust/ or Zed from https://zed.dev/.

3 Check your setup

If Rust is already installed, run `rustup update` first. We will use Rust edition 2024, so any recent stable version will work.

Run these commands:

```
rustc --version
cargo --version
cargo new hello && cd hello && cargo run
```

The last command should print `Hello, world!`

4 If the installation does not work

You can use the Rust Playground at https://play.rust-lang.org/. It runs in your browser and is enough for all the exercises. Paste in the code and click Run.

5 Warm-up before the tutorial

Please spend about 15 to 30 minutes looking at basic Rust syntax. You should know how `let` and `mut` work and have seen functions, structs, `Vec`, and `String`. We will not spend much time on these basics during the tutorial.

Choose whichever resource works best for you:

- For a quick overview, read *Learn X in Y Minutes: Rust* at https://learnxinyminutes.com/rust/.
- For a fuller introduction, read chapters 1 to 3 of *The Rust Programming Language* at https://doc.rust-lang.org/book/.
- For more examples, see *Rust by Example* at https://doc.rust-lang.org/rust-by-example/.
- For extra practice, try Rustlings at https://rustlings.rust-lang.org/.

Chapter 4 of the Rust book covers ownership, which is the topic of our tutorial. You may read it beforehand, but you do not need to. It is available at https://doc.rust-lang.org/book/ch04-00-understanding-ownership.html.

6 Get the tutorial files

Before the tutorial, clone or download the starter code from https://github.com/thecodinglab/rust-ownership-tutorial. The files we will use are:

```
code/python/bank.py      # the Python program we start from
code/rust/               # the first Rust version
```

Open a terminal in `code/rust/` and run `cargo run`. The program does not compile yet. That is intentional, and we will fix it together.